TABLE S6b

MACRO-F1 AND PREDICTION COVERAGE

Clustering	Model	Means macro-F1	Means + fractions macro-F1	Means coverage (%)	Means + fractions coverage (%)
Leiden	GPT-6 Astra	0.4582	0.4490	100.00	100.00
FlowSOM	GPT-6 Astra	0.3028	0.2713	100.00	100.00
SpatialSort	GPT-6 Astra	0.2152	0.2074	100.00	100.00
PIXIE	GPT-6 Astra	0.2777	0.2553	100.00	100.00
Leiden	Claude Opus 5.5	0.3842	0.4165	100.00	100.00
FlowSOM	Claude Opus 5.5	0.2884	0.2999	100.00	100.00
SpatialSort	Claude Opus 5.5	0.1966	0.2226	100.00	100.00
PIXIE	Claude Opus 5.5	0.2527	0.2751	100.00	100.00
Leiden	Gemini 4 Argon	0.4073	0.4133	100.00	100.00
FlowSOM	Gemini 4 Argon	0.3189	0.2917	100.00	100.00
SpatialSort	Gemini 4 Argon	0.2090	0.2154	100.00	100.00
PIXIE	Gemini 4 Argon	0.2689	0.2619	100.00	100.00
Leiden	DeepSeek V4.1 Flash 1017	0.4084	0.2974	100.00	100.00
FlowSOM	DeepSeek V4.1 Flash 1017	0.2373	0.2196	100.00	100.00
SpatialSort	DeepSeek V4.1 Flash 1017	0.1230	0.1145	100.00	100.00
PIXIE	DeepSeek V4.1 Flash 1017	0.1895	0.1660	100.00	100.00

Completed full-arm bank; equal weighting of two test regions; fixed 28-label vocabulary including Noise. One accepted annotation per cluster and condition; no repeated-call SD.